







There are restrictions we want to keep



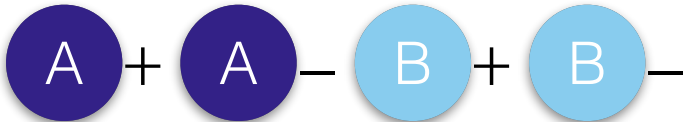
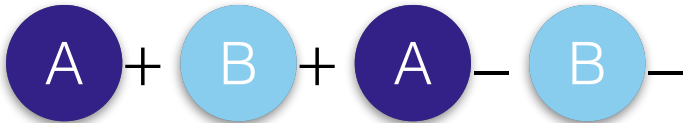
```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    } } }  
}
```



Perminituted



upd(+ -), log(+ -)

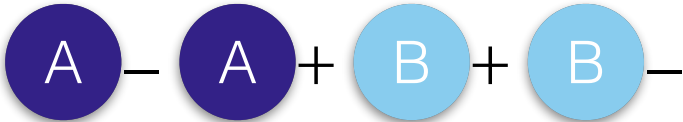


order

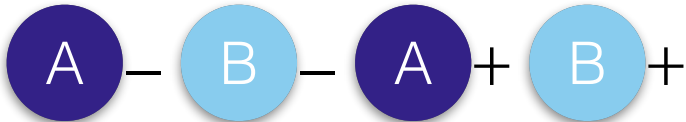
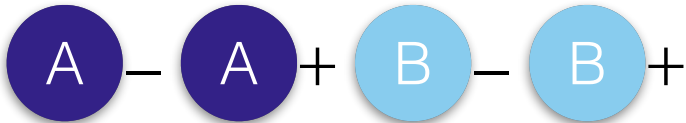
$\text{upd}(+-), \log(-+)$



upd(-+), log(+ -)



upd(-+), log(-+)















In this case, the semantics provide the appropriate restrictions.

# There are restrictions we want to keep

```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    }  
  }  
}
```

In this case, the semantics provide the appropriate restrictions.

|           | upd(+ -), log(+ -)                                              | upd(+ -), log(- +)                    | upd(- +), log(+ -)                    | upd(- +), log(- +)                                              |
|-----------|-----------------------------------------------------------------|---------------------------------------|---------------------------------------|-----------------------------------------------------------------|
| Order     | <div><div>A + B + A - B -</div><div>A + A - B + B -</div></div> | <div><div>A + A - B - B +</div></div> | <div><div>A - A + B + B -</div></div> | <div><div>A - A + B - B +</div><div>A - B - A + B +</div></div> |
| Permitted | ✓                                                               | ✗                                     | ✗                                     | ✓                                                               |



# Insight

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